

Muhammad Hussnain

+92 3186448059 | hussnainshafiq138@gmail.com | linkedin | Github

EXPERIENCE

AI Engineer

May 2026 – Present

Journal Post Group · Lahore, Pakistan

Onsite

- Enhanced LLM-based automation applications using Python, LangChain, and FastAPI, cutting manual workflow effort by 40% across core business processes.
- Architected agentic AI workflows and RAG systems integrating real-time API data sources, achieving measurably improved response accuracy in production.
- Refined prompt engineering and inference pipelines, reducing response latency by ~30% while maintaining output quality across deployed models.
- Partnered with cross-functional teams of 5+ engineers to ship scalable, production-ready AI solutions for business automation use cases.

AI/ML Intern

January 2026 – May 2026

Samsung Innovation Campus – Knowledge Stream · Lahore, Pakistan

Onsite

- Performed EDA, preprocessing, and visualization using NumPy and Pandas across 10+ structured and unstructured datasets, surfacing key data quality issues.
- Implemented and compared supervised and unsupervised models (SVM, KNN, Decision Trees, K-Means clustering) achieving best-in-class accuracy on 3 datasets.
- Constructed and evaluated deep learning architectures in TensorFlow and Keras, improving predictive accuracy by 15%+ through iterative tuning.
- Delivered 2 end-to-end ML projects from raw data ingestion through model deployment, collaborating in agile team workflows.

AI/ML Intern

June 2025 – September 2025

Cross Media Sol · Lahore, Pakistan

Onsite

- Trained and evaluated ML/DL models using Python, TensorFlow, PyTorch, and Scikit-learn to solve real-world business automation challenges.
- Built end-to-end data pipelines handling cleaning, preprocessing, and feature engineering for datasets of varying structure and scale.
- Boosted model accuracy and inference speed through systematic hyperparameter tuning, cutting inference time by ~20% on key production models.

MERN Stack Intern

August 2024 – November 2024

Buggy · Lahore, Pakistan

Online

- collaborating closely with the core engineering team.
- Built and integrated RESTful APIs with Express.js and Node.js, handling authentication, data validation, and CRUD operations.
- Designed responsive, component-based UIs in React, improving usability and reducing frontend bugs through iterative testing.

PROJECTS

Instagram & Facebook Reels Outreach Dashboard | Python, FastAPI, yt-dlp, Groq API, Vanilla JS

- Automated creator outreach research pipeline for Instagram and Facebook Reels, reducing per-reel processing time by ~94% (from 7 minutes manual research to 10 reels in 4 min 20 sec automated).
- Engineered session-based multi-user isolation with a sequential download queue, supporting 9 concurrent users without triggering platform rate-limits.
- Achieved 99% accuracy in automated duplicate detection and creator data extraction, eliminating manual review overhead.
- Optimized per-run processing latency to 6-7 seconds through async task handling and ephemeral disk cleanup, deployed on Render with health-check monitoring for uptime.

AI Agent Studio | *Python, LangGraph, LangChain, FastAPI, Streamlit, Groq, OpenAI, Tavily*

- Built a multi-model agentic platform with LangGraph enabling dynamic switching between Groq (Llama 3.3) and OpenAI (GPT-4o-mini) with zero-downtime model routing.
- Integrated autonomous web search via Tavily API using a ReAct architecture for grounded, real-time information retrieval.
- Designed a decoupled full-stack architecture: FastAPI REST backend paired with a Streamlit glassmorphism UI, supporting concurrent user sessions.
- Ensured production-ready AI interactions through Pydantic validation and custom system prompt engineering.

AirWave – Agentic AI Journalist | *Python, LangGraph, LangChain, FastAPI, Streamlit, BrightData, MCP, ElevenLabs*

- Constructed an end-to-end content pipeline scraping Reddit/Web data with BrightData, generating scripts via Claude AI, and converting to audio with ElevenLabs SDK.
- Implemented async rate limiting and exponential-backoff retry logic, reducing scraping failures by ~80% under high-load conditions.
- Delivered audio content via a FastAPI backend supporting Base64 audio streaming through the custom streamlit frontend.
- Leveraged MCP (Model Context Protocol) and LangChain tool adapters for dynamic, context-aware agent tool execution.

Fine-Tuning DeepSeek LLM using LoRA | *Python, HuggingFace, Transformers, LoRA/PEFT, NLP*

- Fine-tuned a large language model using LoRA/PEFT for domain-specific text generation, reducing GPU memory usage by 70% to enable training on consumer-grade hardware.
- Built a complete pipeline covering tokenization, dataset preparation, training, evaluation, and inference.

Agentic AI Chatbot Using RAG | *Python, LangChain, LangGraph, FastAPI, Twilio, Google Gemini, MedGemma*

- Created an agentic AI system using RAG for intelligent decision-making with multi-step workflow orchestration and prompt management.
- Deployed specialized autonomous agents for empathetic counseling and emergency response, integrating Google Gemini and MedGemma via API-based deployment.

AI Model with Vision and Voice | *Python, Streamlit, GROQ API, Whisper STT, gTTS, Computer Vision*

- Expanded a multimodal AI application combining LLM-based image understanding, Whisper speech-to-text, and gTTS voice output for real-time medical Q&A.
- Deployed inference endpoints via REST APIs, optimizing for low latency and high response quality.

TECHNICAL SKILLS

AI & ML Frameworks: TensorFlow, Keras, PyTorch, Scikit-learn, HuggingFace Transformers, Unsloth, LoRA/PEFT, TRL, W&B

LLM & Agentic AI: LangChain, LangGraph, FastAPI, Streamlit, OpenAI API, Anthropic Claude API, Groq, MCP, RAG, Prompt Engineering

Data & ML Tools: NumPy, Pandas, Matplotlib, OpenCV, EDA, Feature Engineering, Hyperparameter Tuning

Languages: Python, SQL, JavaScript, HTML/CSS, Java, Kotlin, C++, C#, PHP

DevOps & Tools: Git, Docker, AWS, VS Code, PyCharm, Google Colab

Speech & Audio: Whisper STT, gTTS, ElevenLabs SDK

EDUCATION

COMSATS University Islamabad, Sahiwal Campus (CUI)
Bachelor of Science in Computer Science – CGPA: 3.7/4.0

Sahiwal, PK
Feb 2022 – Jan 2026

CERTIFICATIONS & TRAINING

Samsung Innovation Campus AI/ML Program
Knowledge Stream Institute

Jan 2026 – Apr 2026